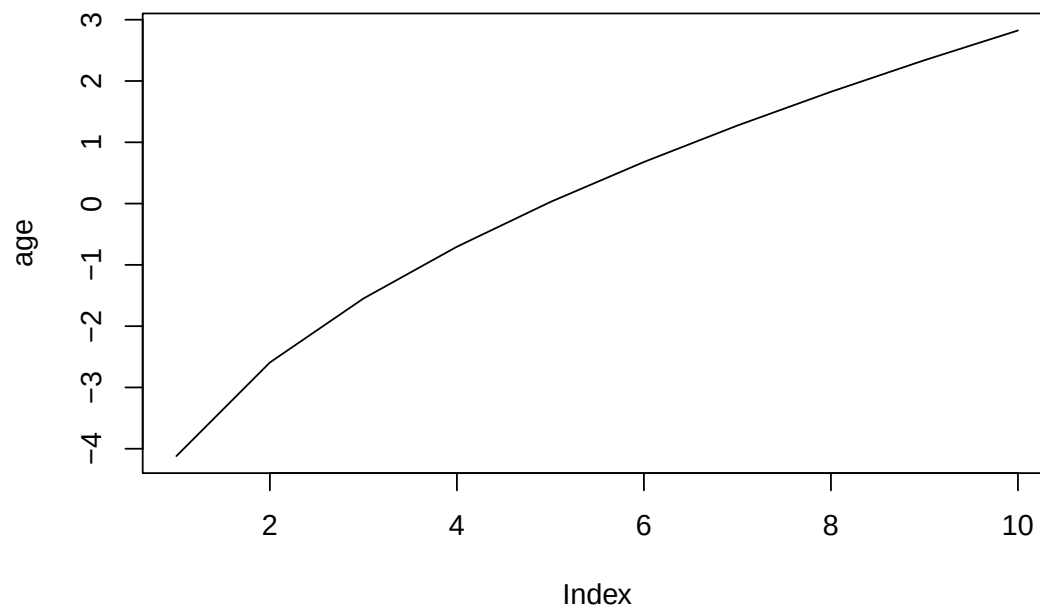


Simulating Age-Period-Cohort Data

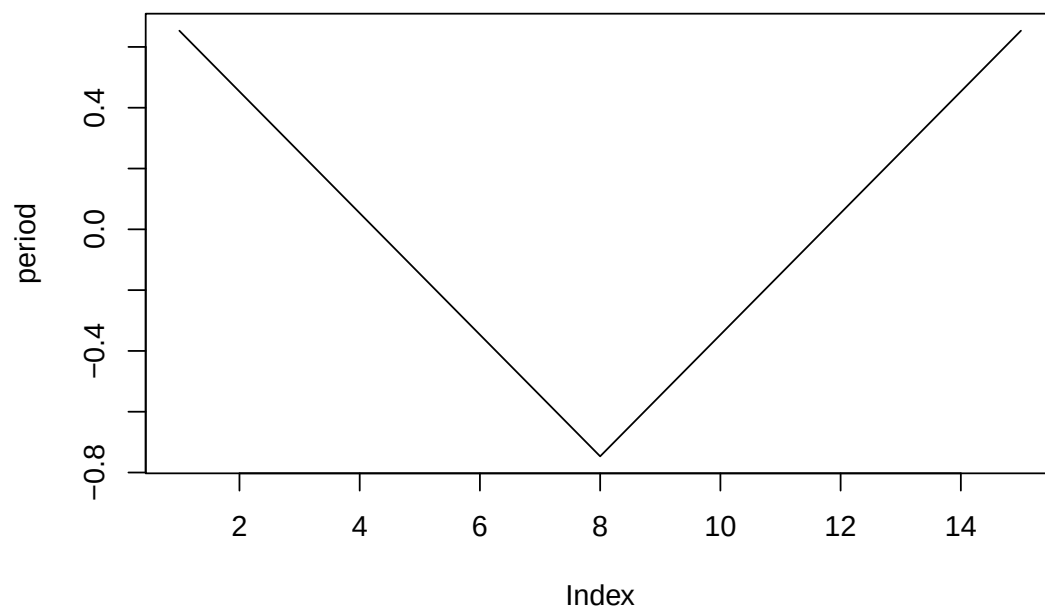
Volker Schmid

2026-06-07

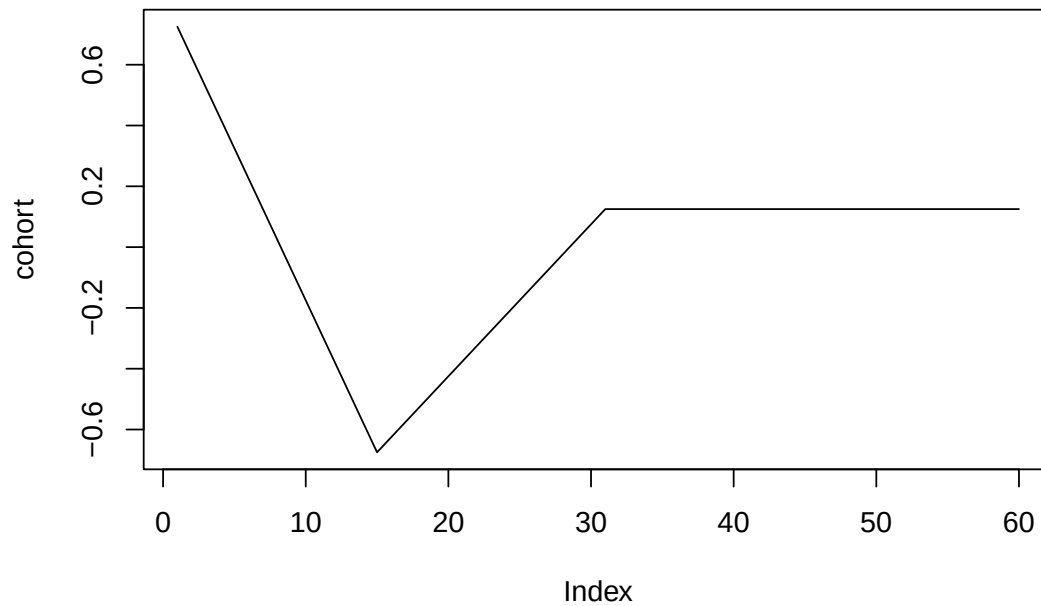
```
age=2*sqrt(seq(1,20,length=10))  
age<- age-mean(age)  
plot(age, type="l")
```



```
period=15:1  
period[8:15]<-8:15  
period<-period/5  
period<-period-mean(period)  
plot(period, type="l")
```



```
periods_per_agegroup=5
number_of_cohorts <- periods_per_agegroup*(10-1)+15
cohort<-rep(0,60)
cohort[1:15]<-(14:0)
cohort[16:30]<- (1:15)/2
cohort[31:60]<- 8
cohort<-cohort/10
cohort<-cohort-mean(cohort)
plot(cohort, type="l")
```



```
simdata<-apcSimulate(-10, age, period, cohort, periods_per_agegroup, 1e6)
print(simdata$cases)
```

```
##      [,1] [,2] [,3] [,4] [,5] [,6] [,7] [,8] [,9] [,10]
## [1,]   3   4  15  49  76 114 158 388 1170 3135
## [2,]   2   2  25  49  75 109 136 328 877 2263
## [3,]   1   5  15  35  59  99 135 238 613 1646
## [4,]   0   5  18  27  49  88  98 164 470 1228
## [5,]   1   8   6  21  35  73  96 136 360  958
## [6,]   1   5   4  19  37  50  73 108 237  633
## [7,]   0   3   6  16  20  44  65  94 175  508
## [8,]   1   0   3  11  30  44  55  70 149  361
## [9,]   0   6   5  17  25  52  64  94 153  460
## [10,]  1   1   5  18  36  86 116 107 174  422
## [11,]  1   3   5  26  52  75 121 172 197  515
## [12,]  0   4   9  24  65 107 146 207 306  582
## [13,]  2   5  17  35  80 137 201 255 409  600
## [14,]  0   3  10  39  84 177 239 341 479  718
## [15,]  0  10  16  51 104 217 322 478 585  694
```

```
simmod <- bamp(cases = simdata$cases, population = simdata$population, age = "rw1",
period = "rw1", cohort = "rw1", periods_per_agegroup =periods_per_agegroup)
```

```
print(simmod)
```

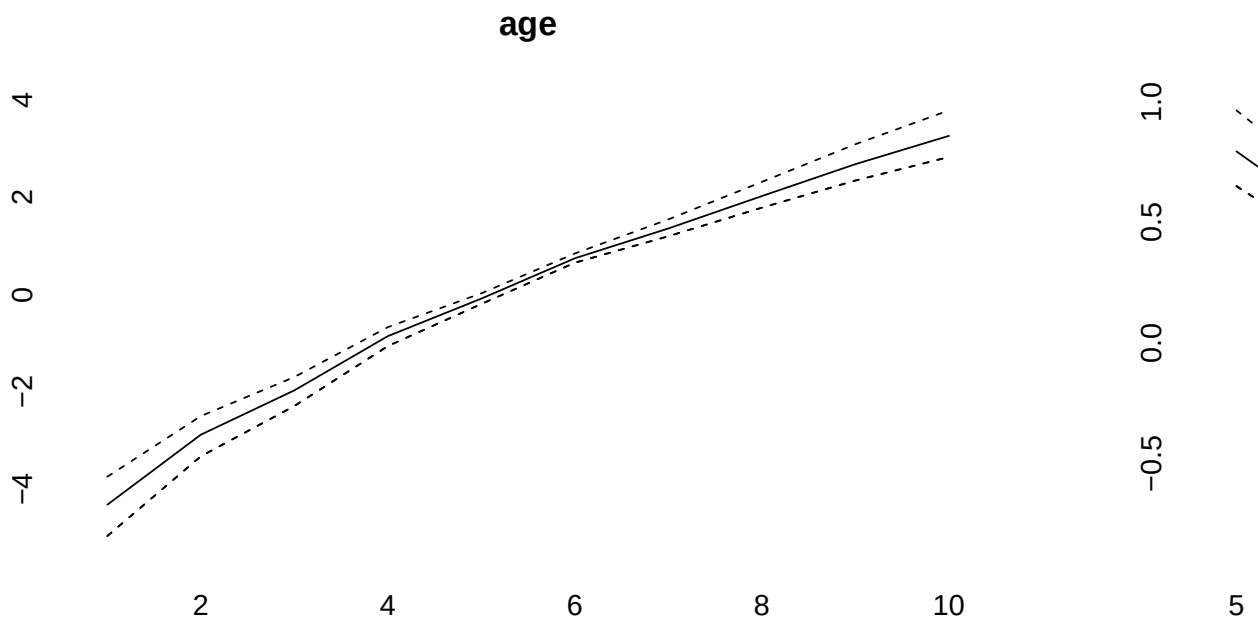
```
##
## Model:
```

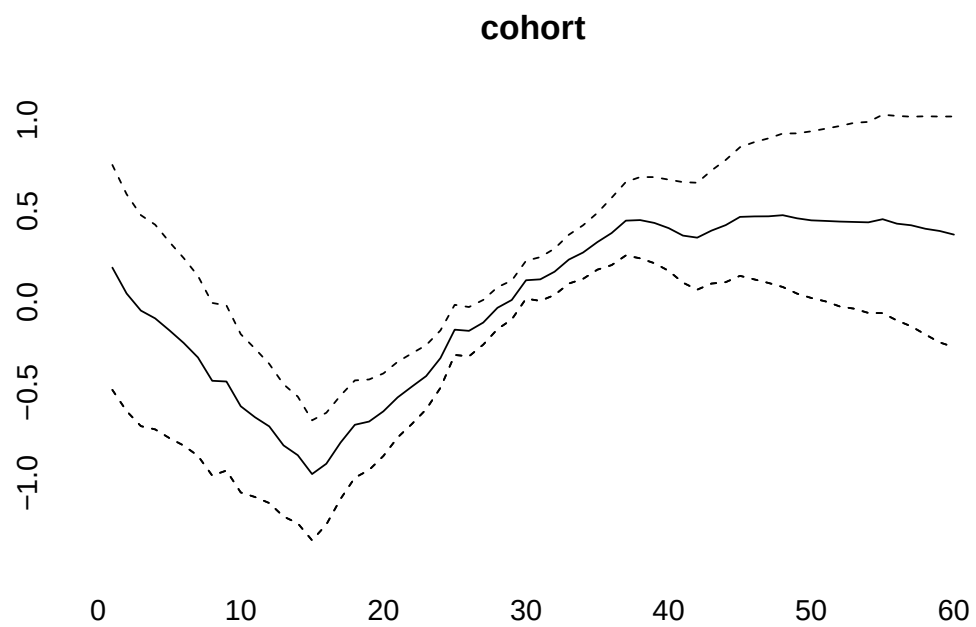
```
## age (rw1) - period (rw1) - cohort (rw1) model
## Deviance:      153.73
## pD:            49.83
## DIC:           203.55
##
##
## Hyper parameters:          5%          50%          95%
## age                        0.548        1.247        2.522
## period                     13.243       25.728       44.901
## cohort                     72.907      113.581      173.855
##
##
## Markov Chains convergence checked succesfully using Gelman's R (potential scale reduction factor).
```

```
checkConvergence(simmod)
```

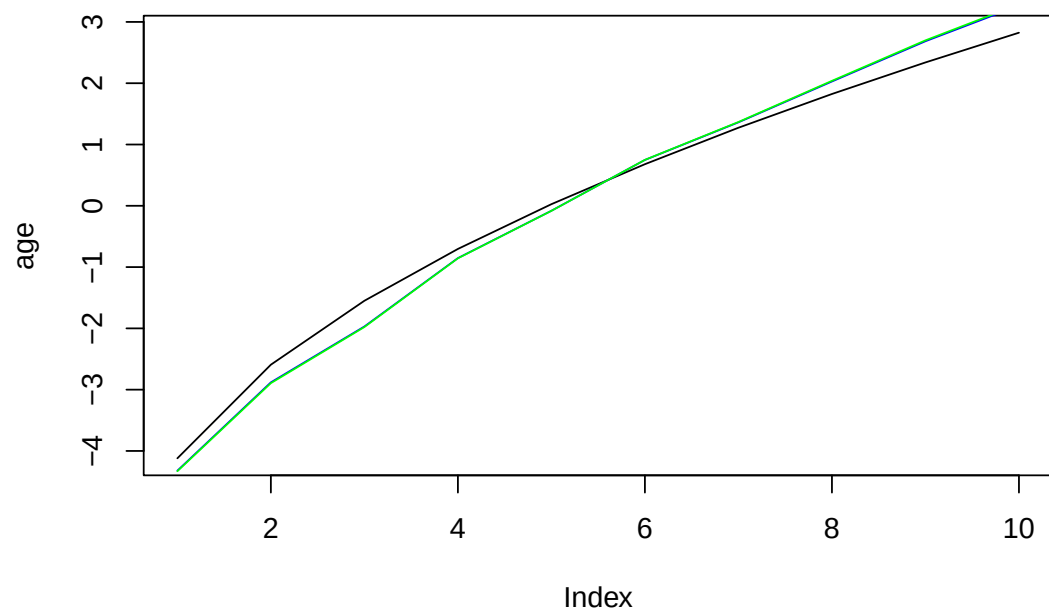
```
## [1] TRUE
```

```
plot(simmod)
```

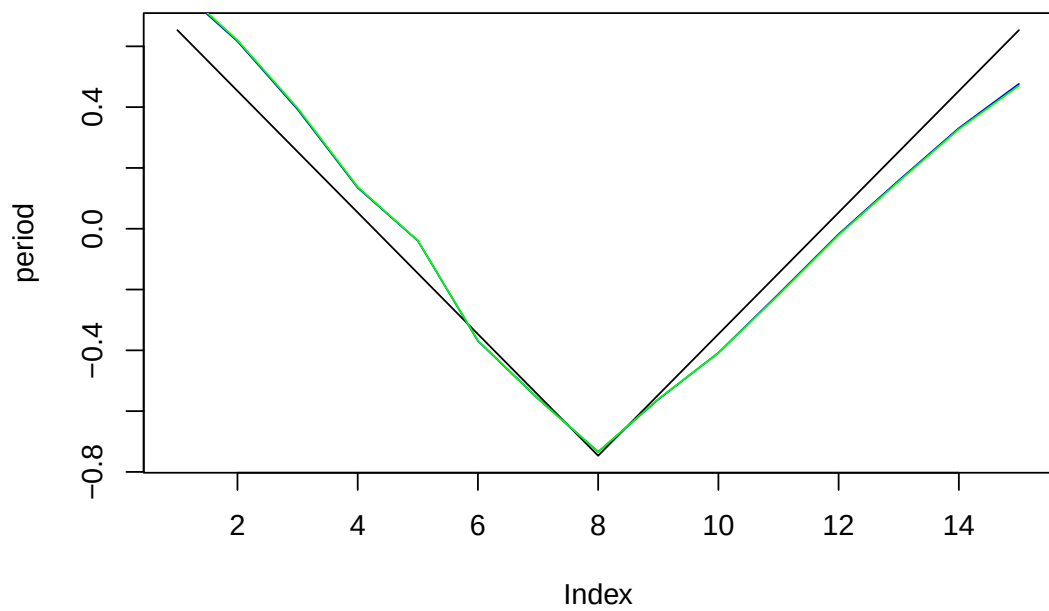




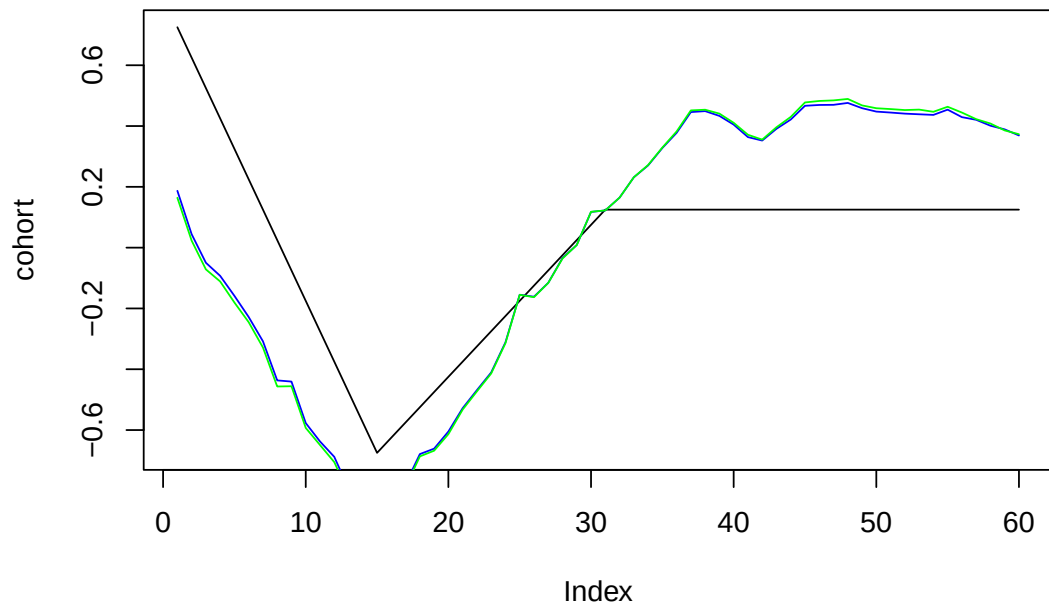
```
effects<-effects(simmod)
effects2<-effects(simmod, mean=TRUE)
#par(mfrow=c(3,1))
plot(age, type="l")
lines(effects$age, col="blue")
lines(effects2$age, col="green")
```



```
plot(period, type="l")
lines(effects$period, col="blue")
lines(effects2$period, col="green")
```

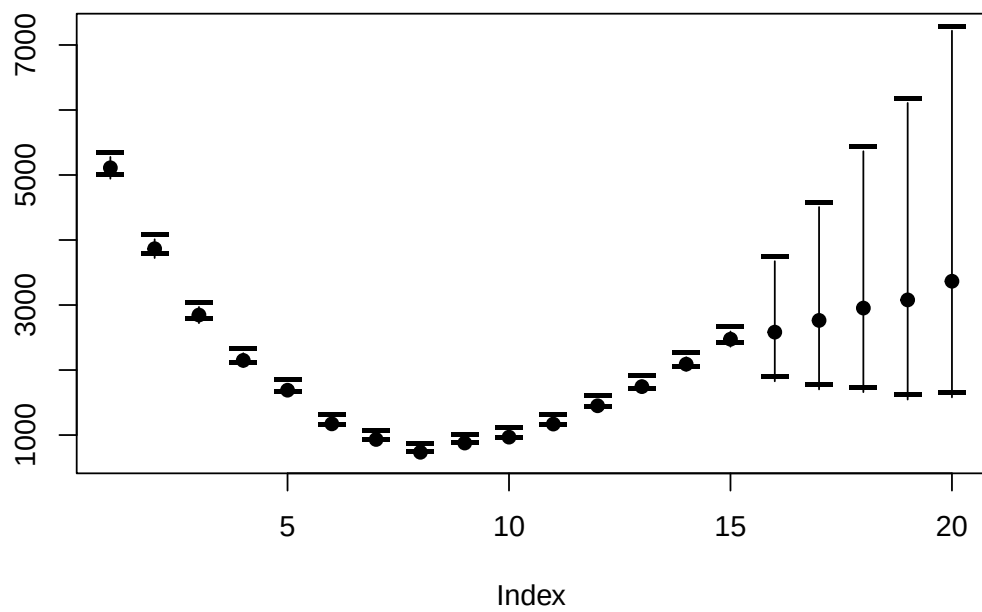


```
plot(cohort, type="l")
lines(effects$cohort, col="blue")
lines(effects2$cohort, col="green")
```

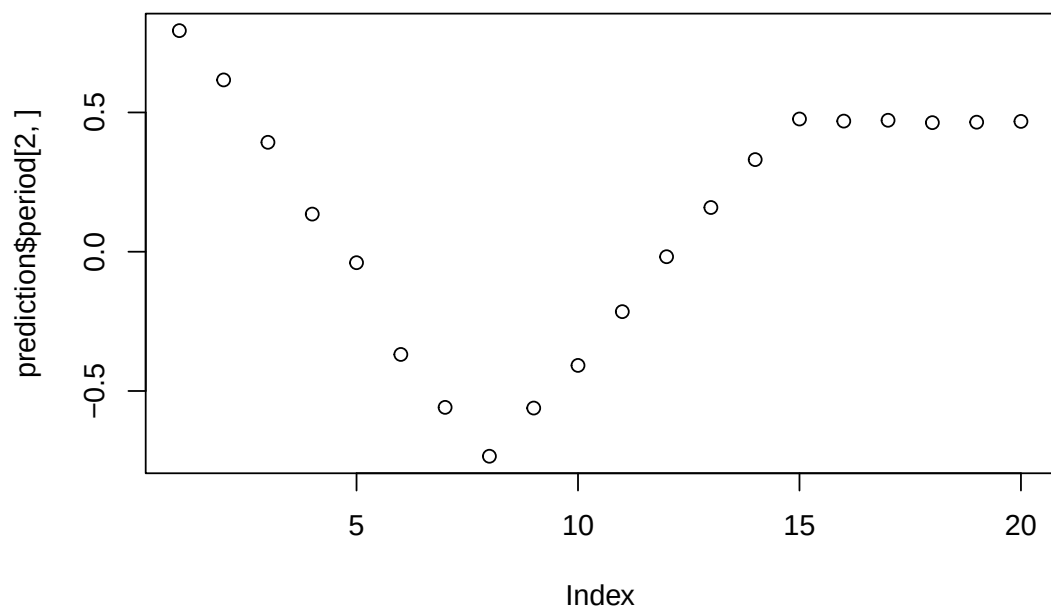


```
prediction<-predict_apc(simmod, periods=5, population=array(1e6,c(20,10)))
```

```
plot(prediction$cases_period[2,], ylim=range(prediction$cases_period),ylab="",pch=19)
points(prediction$cases_period[1,],pch="--",cex=2)
points(prediction$cases_period[3,],pch="--",cex=2)
for (i in 1:20)lines(rep(i,3),prediction$cases_period[,i])
```



```
plot(prediction$period[2,])
```




```
cov_p<-rnorm(15,period,.1)
```

```
simmod2 <- bamp(cases = simdata$cases, population = simdata$population, age = "rw1",  
period = "rw1", cohort = "rw1", periods_per_agegroup = periods_per_agegroup,  
period_covariate = cov_p)
```

```
print(simmod2)
```

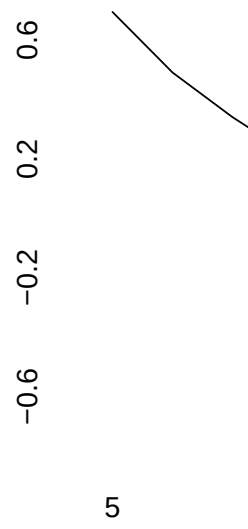
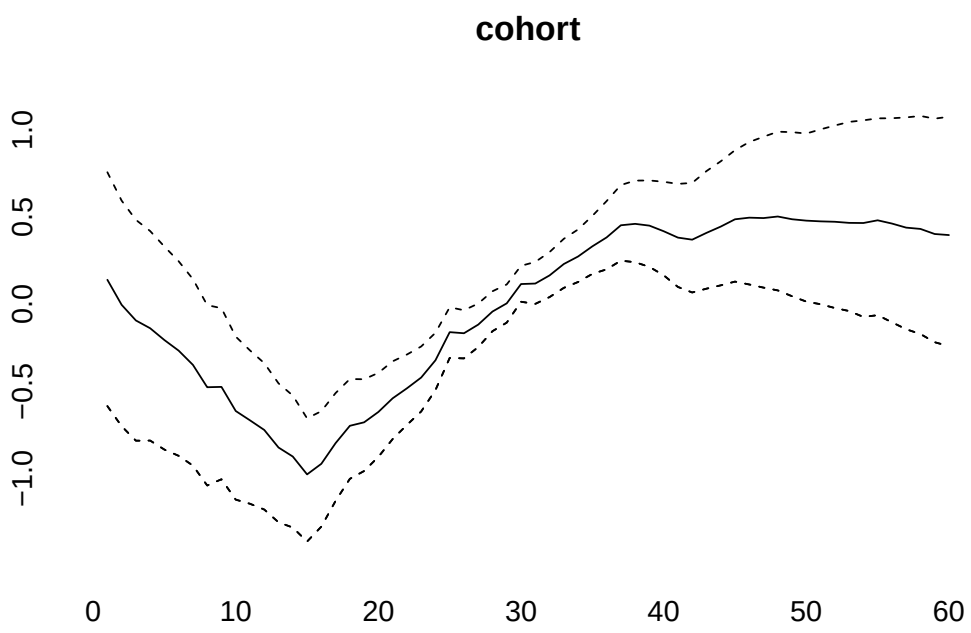
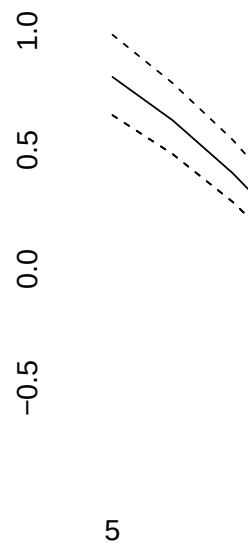
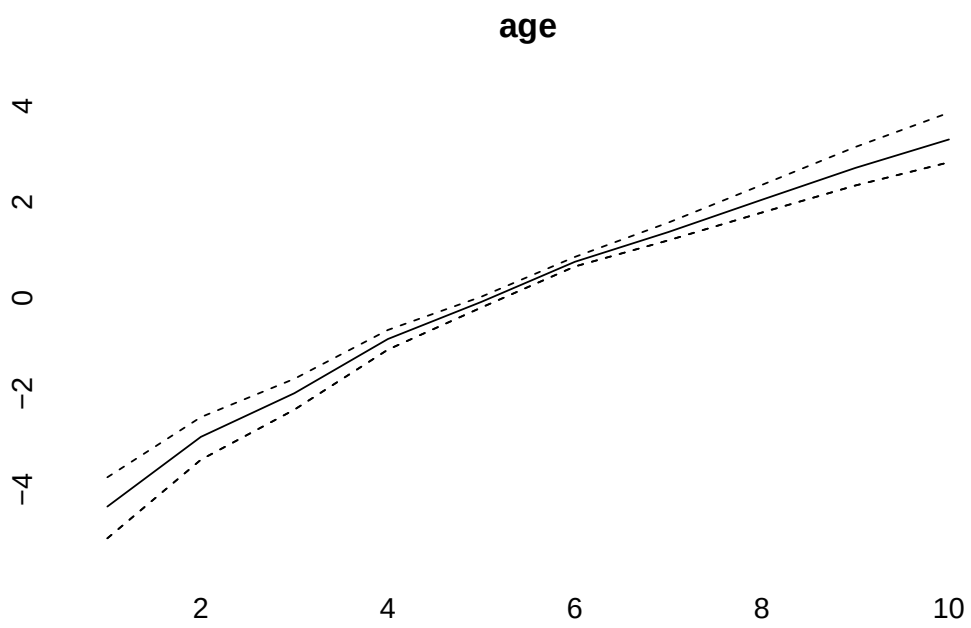
```
##  
## WARNING! Markov Chains have apparently not converged! DO NOT TRUST THIS MODEL!  
##  
## Model:  
## age (rw1) - period (rw1) - cohort (rw1) model  
## Deviance:      153.75  
## pD:            49.85  
## DIC:           203.60  
##  
##  
## Hyper parameters:           5%           50%           95%  
## age                        0.534        1.246        2.505  
## period                     12.700        25.293        43.959  
## cohort                     71.599       113.538       172.393
```

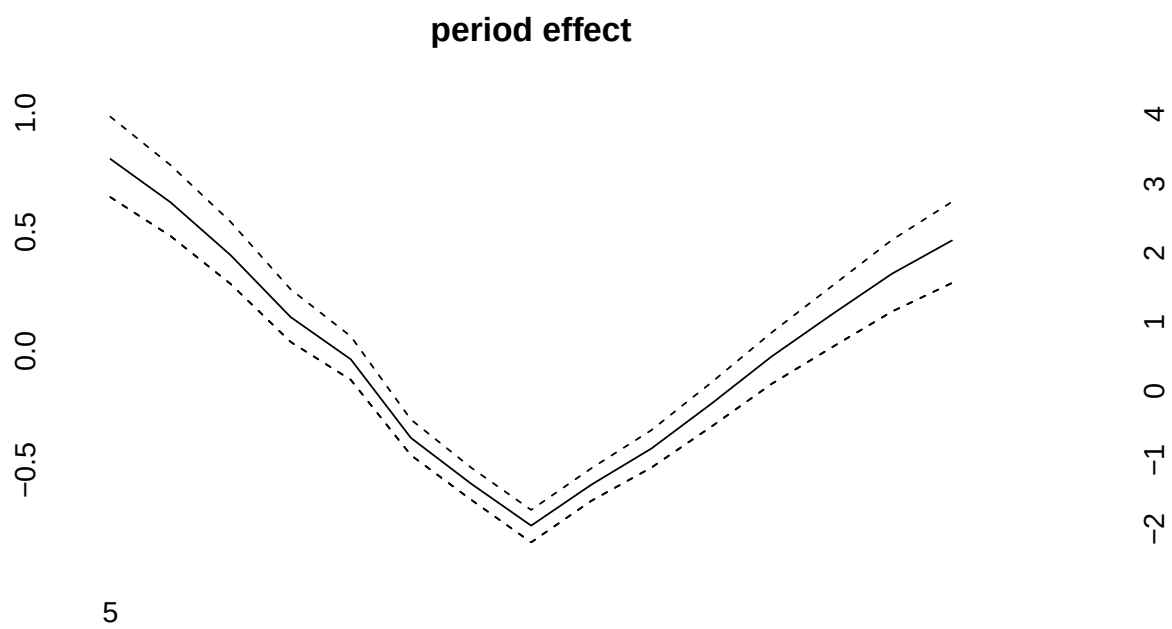
```
checkConvergence(simmod2)
```

```
## Warning: MCMC chains did not converge!
```

```
## [1] FALSE
```

```
plot(simmod2)
```





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